

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Experiment No.: 3 Name: Vansh Mahant PRN: 24070521038 Section: A Batch: A1

1. Experiment Title:

Student Grade Calculator — a web page using HTML, CSS and JavaScript conditional statements (if / else if) to evaluate a student's grade based on marks entered, with live result feedback as the user types.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

3. Experiment Program Code:

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Grade Calculator - Javascript Assignment</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <div class="container">
    <h1>Student Grade Calculator</h1>
    <p class="intro">Javascript conditional statement case study for evaluating student
grades.</p>

    <!-- Calculator Section -->
    <div class="card">
      <h3>Enter Student Marks</h3>
      <form id="studentForm">
        <div class="form-group">
          <label for="name">Student Name:</label>
          <input type="text" id="name" class="form-control" placeholder="Enter full name"
required>
        </div>

        <div class="form-group">
          <label for="marks">Marks (0 - 100):</label>
          <input type="number" id="marks" class="form-control" min="0" max="100"
placeholder="e.g. 78" required>
        </div>

        <button type="submit" class="btn">Calculate Grade</button>
      </form>

      <!-- Result output -->
      <div class="result-box" id="resultBox">
        <div style="font-weight: bold; color: #555;">Result:</div>
        <div class="result-grade" id="gradeOutput">Grade: B</div>
        <div style="font-size: 13px; color: #666; margin-top: 5px;" id="marksOutput">Marks:
78</div>
      </div>
    </div>
  </div>
</div>
```

```
<script src="script.js"></script>
</body>
</html>
```

[styles.css](#)

```
/* Clean, simple student project CSS */
body {
  font-family: Arial, sans-serif;
  background-color: #f4f6f9;
  color: #333;
  margin: 0;
  padding: 20px;
}

.container {
  max-width: 600px;
  margin: 0 auto;
  background-color: #fff;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}

h1 {
  text-align: center;
  color: #2c3e50;
  margin-bottom: 20px;
}

.intro {
  text-align: center;
  color: #7f8c8d;
  margin-bottom: 30px;
}

/* Card Styling */
.card {
  border: 1px solid #ddd;
  border-radius: 6px;
  padding: 20px;
  background-color: #fafafa;
  margin-bottom: 20px;
}

.card h3 {
  margin-top: 0;
  color: #2980b9;
  border-bottom: 2px solid #3498db;
  padding-bottom: 8px;
}

/* Forms */
.form-group {
  margin-bottom: 15px;
}

.form-group label {
  display: block;
  font-weight: bold;
  margin-bottom: 5px;
  color: #34495e;
}

.form-control {
  width: 100%;
  padding: 8px 10px;
  border: 1px solid #ccc;
  border-radius: 4px;
}
```

```

    box-sizing: border-box;
    font-size: 14px;
}

.form-control:focus {
    outline: none;
    border-color: #3498db;
}

/* Buttons */
.btn {
    display: inline-block;
    width: 100%;
    padding: 10px;
    background-color: #3498db;
    color: #fff;
    border: none;
    border-radius: 4px;
    font-size: 16px;
    cursor: pointer;
    font-weight: bold;
    transition: background-color 0.2s;
}

.btn:hover {
    background-color: #2980b9;
}

.btn-secondary {
    background-color: #95a5a6;
    margin-top: 8px;
}

.btn-secondary:hover {
    background-color: #7f8c8d;
}

/* Result box */
.result-box {
    margin-top: 15px;
    padding: 15px;
    background-color: #ebf5fb;
    border-left: 5px solid #3498db;
    border-radius: 4px;
    display: none;
}

.result-box.show {
    display: block;
}

.result-grade {
    font-size: 24px;
    font-weight: bold;
    color: #2c3e50;
    margin-top: 5px;
}

/* Table */
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
}

th, td {
    padding: 10px;
    border: 1px solid #ddd;
    text-align: left;
}

```

```

}

th {
  background-color: #ecf0f1;
  color: #2c3e50;
}

tr:hover {
  background-color: #f9f9f9;
}

.badge {
  padding: 3px 8px;
  border-radius: 3px;
  font-weight: bold;
  font-size: 12px;
}

.badge-aplus { background-color: #2ecc71; color: white; }
.badge-a { background-color: #f1c40f; color: #333; }
.badge-b { background-color: #3498db; color: white; }
.badge-c { background-color: #9b59b6; color: white; }
.badge-fail { background-color: #e74c3c; color: white; }

.delete-btn {
  background-color: #e74c3c;
  color: white;
  border: none;
  padding: 4px 8px;
  border-radius: 3px;
  cursor: pointer;
}

.delete-btn:hover {
  background-color: #c0392b;
}

```

script.js

```

// Document ready
document.addEventListener("DOMContentLoaded", () => {
  // Real-time visual feedback as user types marks
  document.getElementById("marks").addEventListener("input", updateResult);
});

// Run calculation using conditional statements
function calculateGrade(marks) {
  let grade = "";

  // Conditional statements matching logic rules
  if (marks > 90) {
    grade = "A+";
  } else if (marks >= 80 && marks <= 90) {
    grade = "A";
  } else if (marks >= 66 && marks <= 80) {
    grade = "B";
  } else if (marks >= 50 && marks <= 65) {
    grade = "C";
  } else {
    grade = "Fail";
  }

  return grade;
}

// Live feedback update helper
function updateResult() {
  const marksInput = document.getElementById("marks").value;
  const resultBox = document.getElementById("resultBox");
}

```

```

if (marksInput === "" || isNaN(marksInput)) {
    resultBox.classList.remove("show");
    return;
}

const marks = parseFloat(marksInput);
if (marks < 0 || marks > 100) {
    resultBox.classList.remove("show");
    return;
}

const grade = calculateGrade(marks);

// Show result box with values
document.getElementById("gradeOutput").textContent = "Grade: " + grade;
document.getElementById("marksOutput").textContent = "Marks: " + marks;
resultBox.classList.add("show");
}

// Submit handler to trigger final calculation
document.getElementById("studentForm").addEventListener("submit", (e) => {
    e.preventDefault();

    const marks = parseFloat(document.getElementById("marks").value);

    if (marks < 0 || marks > 100) {
        alert("Marks must be between 0 and 100.");
        return;
    }

    updateResult();
});

```

4. Output:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical3/index.html

Student Grade Calculator

Javascript conditional statement case study for evaluating student grades.

Enter Student Marks

Student Name:

Marks (0 - 100):

Calculate Grade

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Student Grade Calculator

Javascript conditional statement case study for evaluating student grades.

Enter Student Marks

Student Name:

Vansh Mahant

Marks (0 - 100):

78

Calculate Grade

Result:

Grade: B

Marks: 78

5. Result/Conclusion:

A Student Grade Calculator web page was successfully designed and implemented using HTML, CSS and JavaScript. The program uses conditional (if / else if) statements to evaluate the grade corresponding to marks entered by the user, with the result updating live as the marks are typed and re-validated on form submission. This experiment reinforced the concepts of conditional logic, event handling, and real-time DOM updates in JavaScript.

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 3_24070521038_Vansh Mahant_A_A1